



Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous College Affiliated to University of Mumbai)

Re-
~~End Semester~~ Examination
~~November 2018~~

Max. Marks: 100

Class: S.E.

Course Code: IT44/CE43

Name of the Course: Operating Systems

Jan-2019

Duration: 180 Minutes

Semester: IV

Branch: IT/COMPS

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks	CO																									
Q 1 (a)	Explain fork system call with example.	5	1																									
Q 1 (b)	What is batch O.S. Explain its working.	5	1																									
Q 1(c)	Explain RAID 0, RAID 1, RAID 5, RAID 6.	10	2																									
Q2(a)	Explain Process state transition diagram with all transitions.	10	2																									
Q2(b)	Solve the following using FCFS and NonPreemptive SJF scheduling algorithm. <table border="1"><tr><td>Process</td><td>Burst time</td></tr><tr><td>P1</td><td>21</td></tr><tr><td>P2</td><td>3</td></tr><tr><td>P3</td><td>6</td></tr><tr><td>P4</td><td>2</td></tr></table> OR Solve the following using Non Preemptive Priority andRound Robin scheduling algorithm. Time quantum=4ms and lower number indicates higher priority. <table border="1"><tr><td>Process</td><td>Burst time</td><td>Priority</td></tr><tr><td>P1</td><td>21</td><td>2</td></tr><tr><td>P2</td><td>3</td><td>1</td></tr><tr><td>P3</td><td>6</td><td>4</td></tr><tr><td>P4</td><td>2</td><td>3</td></tr></table>	Process	Burst time	P1	21	P2	3	P3	6	P4	2	Process	Burst time	Priority	P1	21	2	P2	3	1	P3	6	4	P4	2	3	10	2
Process	Burst time																											
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Q3 (a)	How deadlock can be prevented? Explain how hold and wait, circular wait condition can be used to prevent deadlock.	10	3																									



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Q3 (b)	a. Explain producer consumer problem. b. Consider the methods used by processes P1 and P2 for accessing their critical sections whenever needed, as given below. The initial values of shared boolean variables S1 and S2 are randomly assigned. <table border="1"><tr><td>Method used by P1</td><td>Method used by P2</td></tr><tr><td>While (s1 == s2) ; Critical section s1 = s2</td><td>While (s1 !=s2) ; Critical section S2 = not (s1);</td></tr></table> Which one of the following statements describes the properties achieved? Justify your answer. (A) Mutual exclusion but not progress (B) Progress but not mutual exclusion (C) Neither mutual exclusion nor progress (D) Both mutual exclusion and progress OR Solve following using Banker's algorithm and check for safe sequence if any. <table border="1"><tr><th>Process</th><th>Max</th><th>Allocation</th><th>Available</th></tr><tr><td></td><td>A B C</td><td>A B C</td><td>A B C</td></tr><tr><td>P0</td><td>0 0 1</td><td>0 0 1</td><td></td></tr><tr><td>P1</td><td>1 7 5</td><td>1 0 0</td><td></td></tr><tr><td>P2</td><td>2 3 5</td><td>1 3 5</td><td></td></tr><tr><td>P3</td><td>0 6 5</td><td>0 6 3</td><td></td></tr><tr><td>Total</td><td></td><td>2 9 9</td><td>1 5 2</td></tr></table> How many resources are there of type (A,B,C)?	Method used by P1	Method used by P2	While (s1 == s2) ; Critical section s1 = s2	While (s1 !=s2) ; Critical section S2 = not (s1);	Process	Max	Allocation	Available		A B C	A B C	A B C	P0	0 0 1	0 0 1		P1	1 7 5	1 0 0		P2	2 3 5	1 3 5		P3	0 6 5	0 6 3		Total		2 9 9	1 5 2	5 5 10	3
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Q4 (a)	Consider a simple paging system with the following parameters 2^{31} bytes of addressable physical memory, page size of 2^{10} bytes, 2^{26} bytes of logical address space. (a) How many bits are in a logical address? (b) How many bytes in a frame? (c) How many bits in the physical address specify the frame? (d) How many entries in the page table? (e) How many bits in each page table entry (assume each page table entry includes a valid/invalid bit). OR Example-2, Let's have a reference string: a, b, c, d, c, a, d, b, e, b, a, b, c, d and the size of the frame be 4. Solve using FIFO and optimal page replacement strategy.	10	4																																

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Q4 (b)	What is TLB? Why it is needed ?explain with neat diagram.	5	4
Q4 (c)	What is segmentation? Why it is needed? What is segment table and what it contains?	5	4
Q5(a)	Cylinder requests are 23, 89, 132, 42, 187 . There are 200 cylinders numbered from 0 – 199. The disk head starts at number 100. Calculate total head movement and average seek length using FCFS and SSTF disk scheduling algorithm.	10	2
Q5(b)	Explain chained file allocation method with neat diagram. What is the advantage and drawback of this scheme ? How to overcome its drawback ? OR What is Inode? Explain its need with neat diagram.	10	5